الأسئلة المقالية — 12 متقدم · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (36 سؤالاً)

LO21 الاتصال: التعريف والتطبيقات · Continuity: definition and applications

21.1 LO21 · Q1

Complete every part of the question shown in the image. Write your working in the space below.

الأستاذ كامل بشارة 0502837825

Q1 Consider the function: حدد قيم a والتي تجعل الدالة $f(x)$ متصلة على مجالها

$$f(x)=\frac{x^2+x-2}{x+2}$$

a) Determine the domain of $f(x)$ and state the intervals where $f(x)$ is continuous. حدد مجال الدالة وحدد نقاط عدم الاتصال

b) Determine where $f(x)$ has a removable discontinuity. If possible, define a continuous extension function $g(x)$ that is continuous on all of $\mathbb{R}$. حدد القيم التي يكون للدالة عندها انفصال قابل للإزالة ثم اوجد الدالة الموسعة بحيث تكون الدالة متصلة لجميع القيم

a. Complete all parts of the question.

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21.2 LO21 · Q2

Complete every part of the question shown in the image. Write your working in the space below.

Q2 Consider the function: حدد قيم a والتي تجعل الدالة $f(x)$ متصلة على مجالها

$$g(x) = \frac{x^2 - 9}{x - 3}$$

a) Determine the domain of $f(x)$ and state the intervals where $f(x)$ is continuous. حدد مجال الدالة وحدد نقاط عدم الاتصال

b) Determine where $f(x)$ has a removable discontinuity. If possible, define a continuous extension function $g(x)$ that is continuous on all of $\mathbb{R}$. حدد القيم التي يكون للدالة عندها انفصال قابل للإزالة ثم اوجد الدالة الموسعة بحيث تكون الدالة متصلة لجميع القيم

a. Complete all parts of the question.

21.3 LO21 · Q3

Complete every part of the question shown in the image. Write your working in the space below.

Q3	Determine the values of c that make the function $T(x)$ continuous at $x = 10{,}000$	حدد قيم C والتي تجعل الدالة $T(x)$ متصلة على مجالها $x = 10{,}000$

$$T(x) = \begin{cases} 0, & x \le 0 \\ 0.14x, & 0 < x < 10{,}000 \\ c + 0.21x, & 10{,}000 \le x \end{cases}$$

a. Complete all parts of the question.

21.4 LO21 · Q4

Complete every part of the question shown in the image. Write your working in the space below.

Q4	Determine the values of a *and* b that make the function $T(x)$ continuous	حدد قيم $\boldsymbol{a}$ و b التي تجعل الدالة $\boldsymbol{T(x)}$ متصلة على مجالها

$$T(x) = \begin{cases} 0, & x \le 0 \\ a + 0.12x, & 0 < x \le 20{,}000 \\ b + 0.16(x - 20{,}000), & x > 20{,}000 \end{cases}$$

a. Complete all parts of the question.

21.5 LO21 · Q5

Complete every part of the question shown in the image. Write your working in the space below.

Q5	Determine the values of a *and* $\boldsymbol{b}$ that make the function $f(x)$ continuous at $x = 0$	حدد قيم $\boldsymbol{a}$ و $\boldsymbol{b}$ التي تجعل الدالة $\boldsymbol{f(x)}$ متصلة على مجالها

$$f(x) = \begin{cases} \dfrac{2\sin x}{x}, & x < 0 \\ a, & x = 0 \\ b\cos x, & x > 0 \end{cases}$$

a. Complete all parts of the question.

21.6 LO21 · Q6

Complete every part of the question shown in the image. Write your working in the space below.

Q6	Determine the values of a *and* $\boldsymbol{b}$ that make the function $f(x)$ continuous	حدد قيم $\boldsymbol{a}, \boldsymbol{b}$ والتي تجعل الدالة $\boldsymbol{f(x)}$ متصلة على مجالها

$$f(x) = \begin{cases} a(\tan^{-1} x + 2), & x < 0 \\ 2e^{bx} + 1, & 0 \le x \le 3 \\ \ln(x-2) + x^2, & x > 3 \end{cases}$$

a. Complete all parts of the question.

21.7 LO21 · Q7

Complete every part of the question shown in the image. Write your working in the space below.

Q7	Determine the values of a and b that make the function $f(x)$ continuous	حدد قيم a, b والتي تجعل الدالة $f(x)$ متصلة مجالها

$$f(x) = \begin{cases} ae^x + 1, & x < 0 \\ \sin^{-1}\frac{x}{2}, & 0 \le x \le 2 \\ x^2 - x + b, & x > 2 \end{cases}$$

a. Complete all parts of the question.

21.8 LO21 · Q8

Complete every part of the question shown in the image. Write your working in the space below.

Q8	Determine the values of a *and* $\boldsymbol{b}$ that make the function $f(x)$ continuous	حدد قيم $\boldsymbol{a, b}$ والتي تجعل الدالة $\boldsymbol{f(x)}$ متصلة مجالها

$$f(x) = \begin{cases} -2, & x \leq -1 \\ ax - b, & -1 < x < 1 \\ 3, & x \geq 1 \end{cases}$$

a. Complete all parts of the question.

21.9 LO21 · Q9

Complete every part of the question shown in the image. Write your working in the space below.

Q9	Determine the values of a *and* b that make the function $f(x)$ continuous	حدد قيم a, b والتي تجعل الدالة $f(x)$ متصلة مجالها

$$f(x) = \begin{cases} ax + 2b, & x \leq 0 \\ x^2 + 3a - b, & 0 < x \leq 2 \\ 3x - 5, & x > 2 \end{cases}$$

a. Complete all parts of the question.

22.1 LO22 · Q29

Complete every part of the question shown in the image. Write your working in the space below.

Q29	Determine all vertical and slant asymptotes.	حدد جميع المقاربات العمودية والمائلة ل

$$y = \frac{x^3}{4 - x^2}$$

a. Complete all parts of the question.

22.2 LO22 · Q30

Complete every part of the question shown in the image. Write your working in the space below.

Q30	Determine all vertical and slant asymptotes.	حدد جميع المقاربات العمودية والمائلة ل

$$y = \frac{x^2 + 1}{x - 2}$$

a. Complete all parts of the question.

22.3 LO22 · Q31

Complete every part of the question shown in the image. Write your working in the space below.

Q31	Determine all vertical and slant asymptotes.	حدد جميع المقاربات العمودية والمائلة ل

$$y = \frac{x^3}{x^2 + x - 4}$$

a. Complete all parts of the question.

22.4 LO22 · Q32

Complete every part of the question shown in the image. Write your working in the space below.

Q32	Determine all vertical and slant asymptotes.	حدد جميع المقاربات العمودية والمائلة ل

$$y = \frac{x^4}{x^3 + 2}$$

a. Complete all parts of the question.

22.5 LO22 ·

Find the vertical, horizontal or oblique asymptotes and any hole. Show polynomial division or limits.

$f(x) = \frac{x^2+1}{x-1}$

a. Domain and vertical asymptote

b. Division and oblique asymptote

c. Holes

23.1 LO23 · Q17

Complete every part of the question shown in the image. Write your working in the space below.

Q17	Find an equation of the tangent line to the graph of	اوجد معادلة المماس لمنحنى الدالة عند النقطة المعطاة

$$f(x) = (x^2 + 2x)(x^4 + x^2 + 1) \quad at\ x = 0$$

a. Complete all parts of the question.

23.2 LO23 · Q18

Complete every part of the question shown in the image. Write your working in the space below.

Q18	Find an equation of the tangent line to the graph of	اوجد معادلة المماس لمنحنى الدالة عند النقطة المعطاة

$$f(x) = (x^3 + x + 1)(3x^2 + 2x - 1) \quad at\ x = 1$$

a. Complete all parts of the question.

23.3 LO23 · Q19

Complete every part of the question shown in the image. Write your working in the space below.

Q19	find an equation of the tangent line to the graph of	اوجد معادلة المماس لمنحنى الدالة عند النقطة المعطاة

$$f(x) = \frac{x+1}{x+2} \quad at\ x = 0$$

a. Complete all parts of the question.

23.4 LO23 · Q20

Complete every part of the question shown in the image. Write your working in the space below.

Q20	Find an equation of the tangent line to the graph of	اوجد معادلة المماس لمنحنى الدالة عند النقطة المعطاة

$$f(x) = \frac{x+3}{x^2+1} \quad at\ x = 1$$

a. Complete all parts of the question.

23.5 LO23 ·

Use the Product Rule to find f′(x), then find the tangent line at x=1.

$$f(x) = (x^2 + 1)(x^3 - 2)$$

a. Product Rule derivative

b. Tangent line

24.1 LO24 · Q3

Complete every part of the question shown in the image. Write your working in the space below.

Q3	Find the derivative of the function.	اوجد مشتقة الدالة

$$f(t) = \tan^3(2t) - \csc^4(3t)$$

a. Complete all parts of the question.

24.2 LO24 · Q4

Complete every part of the question shown in the image. Write your working in the space below.

Q4	Find the derivative of the function. اوجد مشتقة الدالة

$$f(t) = t^2 + 2\cos^2(4t)$$

a. Complete all parts of the question.

24.3 LO24 · Q5

Complete every part of the question shown in the image. Write your working in the space below.

Q5	Find the derivative of the function.	اوجد مشتقة الدالة

$$f(x) = x\cos(5x^2)$$

a. Complete all parts of the question.

24.4 LO24 · Q6

Complete every part of the question shown in the image. Write your working in the space below.

Q6	Find the derivative of the function.	اوجد مشتقة الدالة

$$f(x) = x^2 \sec(4x)$$

a. Complete all parts of the question.

24.5 LO24 · Q7

Complete every part of the question shown in the image. Write your working in the space below.

Q7	Find the derivative of the function.	اوجد مشتقة الدالة

$$f(x) = \frac{\sin(x^2)}{x^2}$$

a. Complete all parts of the question.

24.6 LO24 · Q8

Complete every part of the question shown in the image. Write your working in the space below.

Q8	Find the derivative of the function. اوجد مشتقة الدالة

$$f(x) = \frac{x^2}{\csc^4(2x)}$$

a. Complete all parts of the question.

25.1 LO25 · Q35

A bacterial population starts at 200 and triples every day. (a) Find P(t), the population after t days. (b) Find the instantaneous percentage rate of change, 100P′(t)/P(t).

a. Complete all parts of the question.

25.2 LO25 · Q36

A bacterial population starts at 500 and triples every day. (a) Find P(t), the population after t days. (b) Find the instantaneous percentage rate of change, $100P'(t)/P(t)$.

a. Complete all parts of the question.

25.3 LO25 · Q37

Complete every part of the question shown in the image. Write your working in the space below.

Q37 The concentration of a certain chemical after t seconds of an autocatalytic reaction is given by the function

يعطى تركيز مادة كيميائية معينة بعد مرور t ثانية من تفاعل التحفيز الذاتي بالدالة التالية:

$$c(t) = \frac{6}{2e^{-6t} + 1}, \quad t \geq 0$$

a) Show that $c'(t) > 0$ for all $t \geq 0$.

بين ان $c'(t) > 0$ لجميع قيم $t \geq 0$

b) Using the results from parts (a) justify that the concentration of the chemical never exceeds 6.

باستخدام النتائج الواردة في الفرعين (أ) و برهن على أن تركيز المادة الكيميائية لا يتجاوز 6 أبداً .

a. Complete all parts of the question.

25.4 LO25 · Q37

For t≥0: (a) Show that c′(t)>0. (b) Show that the concentration never exceeds 5 and find its limit as t→∞.

$$c(t) = \frac{10}{9e^{-10t}+2}$$

a. Complete all parts of the question.

25.5 LO25 ·

A population is modelled by P(t)=400e^(0.3t), with t in days. (a) Find P′(t). (b) Find its instantaneous percentage growth rate. (c) Find the doubling time to two decimal places.

$$P(t) = 400e^{0.3t}$$

a. Derivative

b. Percentage rate

c. Doubling time

26.1 LO26 · Q

Complete every part of the question shown in the image. Write your working in the space below.

Q	Find a value of c satisfying the conclusion of the Mean Value Theorem for اوجد قيمة c التي تحقق شروط نظرية القيمة المتوسطة
1)	$f(x) = x^2 + 1 \quad \text{on } [-2,2].$

a. Complete all parts of the question.

26.2 LO26 · Q

Complete every part of the question shown in the image. Write your working in the space below.

Q	Find a value of c satisfying the conclusion of the Mean Value Theorem for	اوجد قيمة c التي تحقق شروط نظرية القيمة المتوسطة
2)	$f(x) = x^2 + 1 \quad \text{on } [0,2].$	

a. Complete all parts of the question.

26.3 LO26 · Q

Complete every part of the question shown in the image. Write your working in the space below.

Q	Find a value of c satisfying the conclusion of the Mean Value Theorem for	اوجد قيمة c التي تحقق شروط نظرية القيمة المتوسطة
3)	$f(x) = x^3 + x^2$ on $[0,1]$.	

a. Complete all parts of the question.

26.4 LO26 · Q

Complete every part of the question shown in the image. Write your working in the space below.

Q	Find a value of c satisfying the conclusion of the Mean Value Theorem for	اوجد قيمة c التي تحقق شروط نظرية القيمة المتوسطة
4)	$f(x) = x^3 + x^2$ on $[-1,1]$.	

a. Complete all parts of the question.

26.5 LO26 · Q

Complete every part of the question shown in the image. Write your working in the space below.

Q	Find a value of c satisfying the conclusion of the Mean Value Theorem for	اوجد قيمة c التي تحقق شروط نظرية القيمة المتوسطة
5)	$f(x) = \sin x \quad \text{on} \quad \left[0, \frac{\pi}{2}\right].$	

a. Complete all parts of the question.

26.6 LO26 · Q

Complete every part of the question shown in the image. Write your working in the space below.

Q	Find a value of c satisfying the conclusion of the Mean Value Theorem for	اوجد قيمة c التي تحقق شروط نظرية القيمة المتوسطة
6)	$f(x) = \sin x \quad on\ [-\pi, 0\,]$.	

اطيب التمنيات للجميع



a. Complete all parts of the question.